4/7/2026

PetConnect

A Pet Adoption Web Platform

Oliver D'Avino, The Vinh Vuong, James Luciano, Kiara Bartuccio

ECOMMERCE – WINTER 2026

Table of Contents

Team lead ***1***

Team Journal **1**

Project Description ***1***

User Side (Front-End) - Kiara **2**

Admin Side (Back-End) 2

Pet Listings **2**

Search & Filtering System **2**

User Accounts & Security 3

Adoption Request System(client) 3

Adoption History (Admin) 3

Admin Dashboard 3

AI Assistant 3

Internal Design 4

Database design 4

Task Breakdown ***4***

Preparation 4

Front - end Development(users) 5

Back-end Development 5

Front-end Development (Admin) 6

Integration 6

Testing 6

Deployment 6

Teacher: Tiago Bortoletto Vaz

Section - 00001

Kiara Bartuccio - 2393920
The Vinh Vuong - 631128
Oliver Davino - 6252915
James Luciano - 6304894

Team lead

The team leader for deliverable 2, is James Luciano

Repository URL

https://github.com/fatherxcracker/eCommerceProject.git

Team Journal

So far everyone has been more or less equally contributing to the project.
Here is a brief description of everyone's contributions:

- Kiara Bartuccio: Wrote project description,
- Oliver D'Avino: Designed the documentation
- James Luciano: Designed the UML class diagram
- The Vinh Vuong: Designed the figma,

For our meetings, we will be communicating through messages either on Instagram or Discord. These meetings don't necessarily have a set frequency as we communicate whenever we need help from one another or are working on the project throughout the week.

Team leads per area

- Front-end: Kiara Bartuccio
- Back-end: Oliver D'Avino
- Database: James Luciano
- Deployment: The Vinh Vuong

Project Description

The purpose of PetConnect is to enhance the process of pet adoption by introducing a centralized online system in which people can easily find and adopt pets. Most shelters and individuals find it hard to find potential adopters since they have minimal exposure and old systems. This site will focus on raising awareness among more individuals about available pets and providing a more efficient, accessible, and user-friendly adoption process.

Web Application Design, Identification, and description

https://www.figma.com/make/kjEbJaNW8V9MYxzbeZ9IjC/Pet-Adoption-Web-Application?t=KqO1AQWtN76blY2j-1

User Side (Front-End) - Kiara

This is what clients (adopters) will be presented with:

- Homepage
- Pet browsing page
- Pet detail page
- Search and filter system
- User account pages (login/register/profile)
- Adoption request page
- Adoption history page
- AI chat support

Admin Side (Back-End)

- Admin dashboard overview
- Manage pets (add/edit/delete)
- Manage users
- Manage adoption requests
- View system activity
- Adoption history page

Pet Listings

Users will be able to view a list of pets categorized. Each pet will display:

- Name
- Breed
- Age
- Description
- Images
- Status (available or adopted)

Search & Filtering System

Users can search pets using:

- Keywords
- Breed
- Age
- Size
- Location

User Accounts & Security

Users will have:

- Register an account
- Log in securely
- Log out
- Reset passwords
- Password encryption
- Input validation
- Two-factor authentication

Adoption Request System(client)

Users can apply to adopt a pet by submitting a form. Each request will:

- Be stored in the database
- Have a status (pending, approved, rejected)
- Will be visible to both user and admin

Adoption History (Admin)

Users can view pets they have adopted.
Admins can track all previous adoptions.

Admin Dashboard

Admins will:

- Add/edit/remove pets
- Approve or reject adoption requests
- Manage user accounts
- Monitor system activity

AI Assistant

The application will include a simple AI/chat feature that:

- Answers common questions
- Helps users navigate the website

Internal Design

This application will follow an MVC structure:

- **Model:** Handles data (users, pets, requests)
- **View:** Handles UI (pages)
- **Controller:** Handles logic (requests, actions)

Database design

Main entities:

- User
- Pet
- Category
- AdoptionRequest
- AdoptionHistory

Task Breakdown

Database

Responsible: The Vinh
Estimated Completion: April 18, 2026

Database Schema Design: Design all database tables including User, Pet, Category, AdoptionRequest, and AdoptionHistory, and define how they are connected.

Create Tables: Implement the database structure by creating tables, primary keys, and relationships using SQL.

Seed Data: Insert sample data such as pets and categories to allow testing of the application.

Preparation

Responsible: The entire team
Estimated Completion: April 2nd, 2026

- Figure out a project idea: Decide on the main concept of the application and what problem it solves.

- Know the features that we will implement: Identify all core features such as pet listings, search, adoption requests, and admin dashboard.

- Design database: Plan the database structure including tables and relationships.

- Set up GitHub: Create repository and organize project structure for team collaboration.
- Know what we will use to create this project like, which database

Front - end Development(users)

Responsible: Kiara
Estimated Completion: April 18, 2026

- Build homepage: Design and create the main page with navigation and featured pets.
- Create a pet listing page: Display all pets with categories and basic information.
- Create a pet detail page: Show full details of a selected pet including images and description.
- Implement search/filter UI: Create search bar and filtering options for users.
- Build login/register pages: Design user authentication pages for account access.
- Create adoption form UI: Design the form where users submit adoption requests.

Back-end Development

Responsible: James
Estimated Completion: April 30, 2026

- Set up project structure: Organize files using MVC and connect to the database.
- Create database tables: Implement tables for users, pets, categories, and requests.
- Implement authentication system: Develop login, registration, and password security logic.
- Connect front-end to database: Ensure data flows correctly between UI and database.
- Implement pet listing logic: Create backend functions to retrieve and display pets.
- Implement search functionality: Develop backend search and filtering queries.
- Implement adoption request system: Handle storing and updating adoption requests.
- Implement AI: Add a basic AI/chat feature to assist users.

Front-end Development (Admin)

Responsible: Oliver
Estimated Completion: April 31, 2026

- Create admin login: Set up secure access for administrators.
- Build dashboard: Design the admin interface to manage the system.
- Add pet management features: Allow admins to add, edit, and delete pet listings.
- Add user management features: Allow admins to view and manage user accounts.
- Add request approval system: Allow admins to approve or reject adoption requests.

Integration

Responsible: Oliver
Estimated Completion: May 3, 2026

- 2FA (Two-Factor Authentication): Add extra security verification during login.
- Flash messages: Display success/error messages for user actions.
- Chat box: Implement a chatbot interface for user assistance.

Testing

Responsible: Kiara
Estimated Completion: May 5, 2026

- Test all the functions: Check all features (login, search, adoption, admin tools).
- Fix bugs if needed: Resolve issues found during testing.
- Improve performance if needed: Optimize loading speed and user experience.

Deployment

Responsible: James
Estimated Completion: May 8, 2026

- Upload project to GitHub: Push final version of the project to repository.
- Configure hosting: Set up server and environment variables.
- Deploy application: Make the application live on the hosting platform.
- Perform final testing: Test the live application and fix any remaining issues.

Tools and Technologies

- Figma
- Google Docs
- Word
- Visual Studio code
- Inspo links :
    - https://spcaroussillon.com/en/adopt/
    - https://animatch.ca/
    - https://www.spca.com/en/adoption/